



CS-Lab No. 1

MicroProgramming

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Computer Structure Class Spring 2025

Sharif University of Technology

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I. Objectives

The goal of this experiment is to become familiar with an arithmetic and logic unit (ALU) and build a simple processor using microprogramming, akin to a simple pocket calculator. In the course « Logic Circuits and Digital Systems, » you have seen that the ALU, or Arithmetic Logic Unit, is a processing unit capable of performing arithmetic and logic operations. To use it, the desired operation type must be specified through control signals.

In the circuit designed for this experiment, the control unit consists of an EEPROM that holds the circuit's control bits. In this EEPROM, the 8 control bits of the circuit, which we use as control signals for the system, include the `select` signal for the ALU, `enable output` signals for the registers, and `carry` and `mode` signals for the ALU. These are stored in successive bytes and used at different times. By using a counter IC, we determine which state we are in during each clock cycle, and thus, it makes sense that the output of the counter is used as an address input to the EEPROM.

II. Pre-Lab Activities

II.1. Questions

1. What do you know about an ALU? In your opinion, which operation causes the most propagation delay in the ALU (remember bit propagation in Ripple Carry ADDer)? Study the datasheet of the `74LS181` IC and answer the following two questions:
2. Which inputs determine the ALU's operation? Which output bit of the IC has more delay than the others?
3. Can you generate the output of `A XNOR B` ? If your answer is yes, please explain how to do this.
5. What is the memory size of the EEPROM `AT28C16` ? What is the maximum access time to the contents of this memory in nanoseconds?
6. If we want to read the content of the first 16 memory locations using a counter (IC `74HC161`), how should we configure the hardware?

II.2. Preparations

For this experiment you should install an OpenSource simulation software named `Digital` . In order to do so, you must first install a `JRE` implementation, for which one is conveniently uploaded to your CW page. After this you just extract the contents of the `Digital.zip` archive and run the `Digital.jar` program. We strongly recommend going through the initial tutorial of the program and the documentation in order to get familiar with the user interface. Specifically, We recommend that you check the following components:

- MUX
- EEPROM
- Adder
- Counter
- Clock

- Multi-Input Logic Functions

You may consult the official documentation of the program (available in the `docu` directory where you extracted the archive) in case any question arises.

In order to use components, you should get familiar with their Input/Output signals and their meanings, and there is no better reference for this other than the help button from the properties dialog.

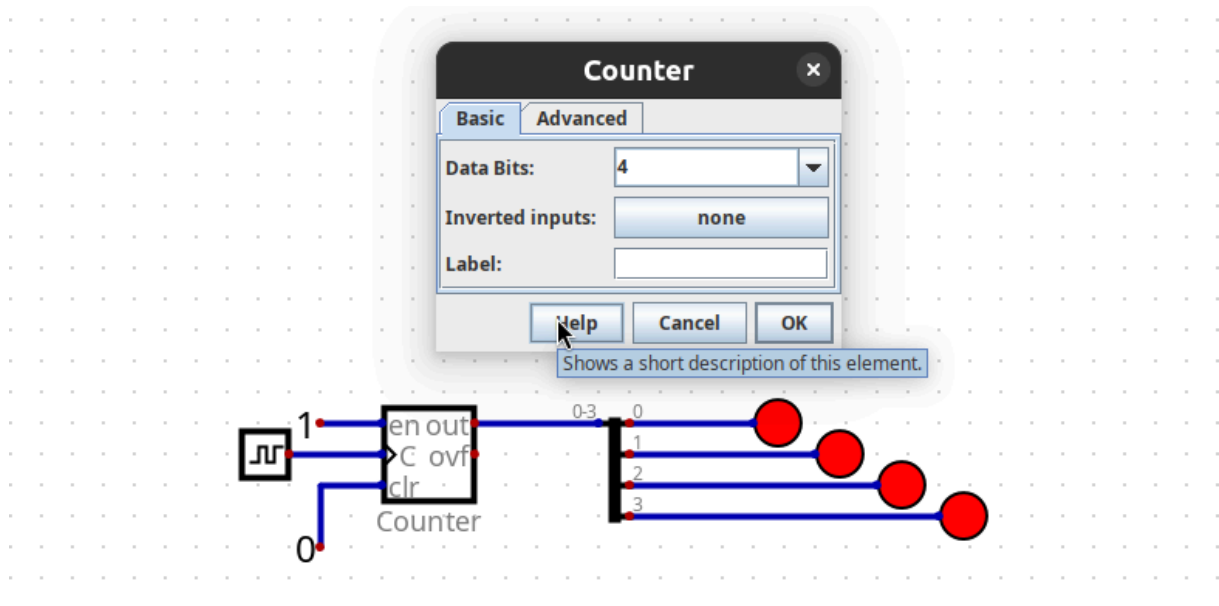


Fig. 1 — You Can Open the Properties Dialog by Right Clicking on the Component

Download Links (Also Available on CW)

- Digital Github [Repository](#)
- JRE Download [Page](#)

II.3. Simulation

Answer the following questions and / or run the simulations. Attach a screenshot of your final circuit.

There is no need to record every step of the simulation. A screenshot of the final circuit along with an Input/Output sample, in addition to a brief explanation will be enough to receive a full score.

S1: XOR From NAND

Create an XOR gate and verify its truth table only using buttons, NAND gate, and an LED. **You can only use NAND gates** to build the XOR gate.

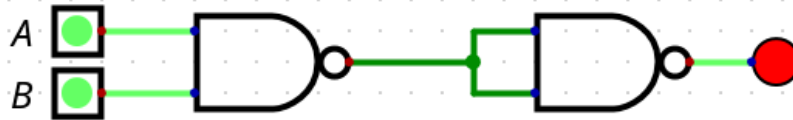


Fig. 2 — Building an AND Gate Only Using Two NAND Gates

S2: Counting Sheep

Using the « Counter » part from the Digital program's library (You can find it in « Components > Memory > Counter ») build a circuit that counts from **0 to 31**.

A « Splitter/Merger » from « Components > Wires > Splitter/Merger » can be used to split the counter's « thick », 5 bit output to 5 « thin » single bit wires.

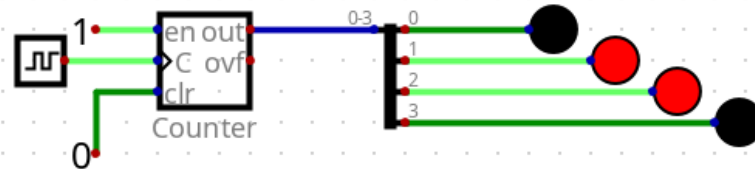


Fig. 3 — A Counter Circuit That Counts from 0 to 15